LX200V30 WisLink 500 Mbps Broadband PLC Module + EVB Datasheet

Overview

Description

LX200V30 is a new type of data transmission product based on OFDM (Orthogonal Frequency Division Multiplexing). The maximum transmission rate of power lines can reach 500 Mbps. Standard Ethernet ports, as well as the two interfaces of twisted pair and coaxial cable, are provided. The module supports 128-bit AES encryption and communicates via a frequency band of 2-68 MHz, and the existing CATV signal or wired broadcasting is not affected. It can be put into work after the wiring is completed without any user configuration. This will advance the product's time-to-market and increase the flexibility of product functionality.

Features

- 500 Mbps PHY rate
- 128-bit encryption
- Range:
  - Power line: up to 300 m
  - Twisted pair: up to 600 m
  - Coaxial cable: up to 2000 m
- Compliant with Homeplug AV specification and IEEE1901 standards
- MII communication interface for communication with any Ethernet-based device
- Data transmission via a coaxial cable, twisted pair, or a power line
- Working frequency range 2-68 MHz
- Dimensions: 40 × 30 mm

Applications

- Smart monitoring instruments
- Energy management systems
- Smart home
- Medical systems
- Automobile electronics charging pile
- Industry

Specifications

Block Diagram

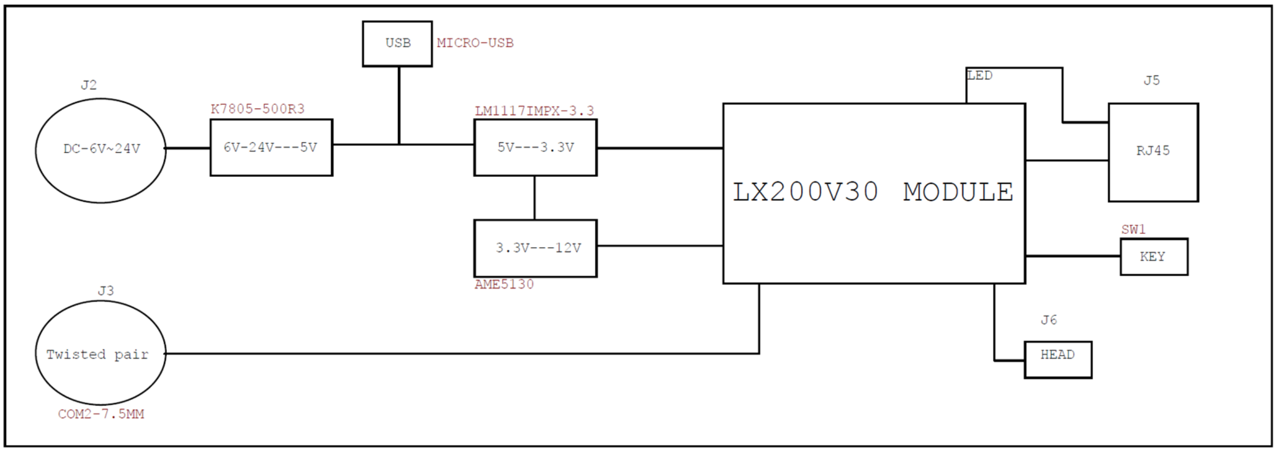

Figure 1: LX200V30 block diagram

Parameters

Parameter	Value
Max. data rate	1000 Mbps
DC interface	4.75 V~36 V
V50 supported unit connection	6~8

Hardware

The hardware specification is categorized into five parts. It shows the interfacing and the pinouts of the board and their corresponding functions and diagrams. It also covers the electrical and mechanical parameters of the LX200V30 WisLink 500 Mbps Broadband PLC Module + EVB.

Interfaces

Evaluation Board (EVB)

The external interface circuits required by the LX200V30 backplane include 7 ~ 28 V, 3.3V BUCK, DC power connector, Ethernet RJ45, PLC coupling transmission transformer, and other coupling parts and their interfaces. The auxiliary interface test board is designed to be convenient for testing. When customers design products, changes will be made according to different structure of products.

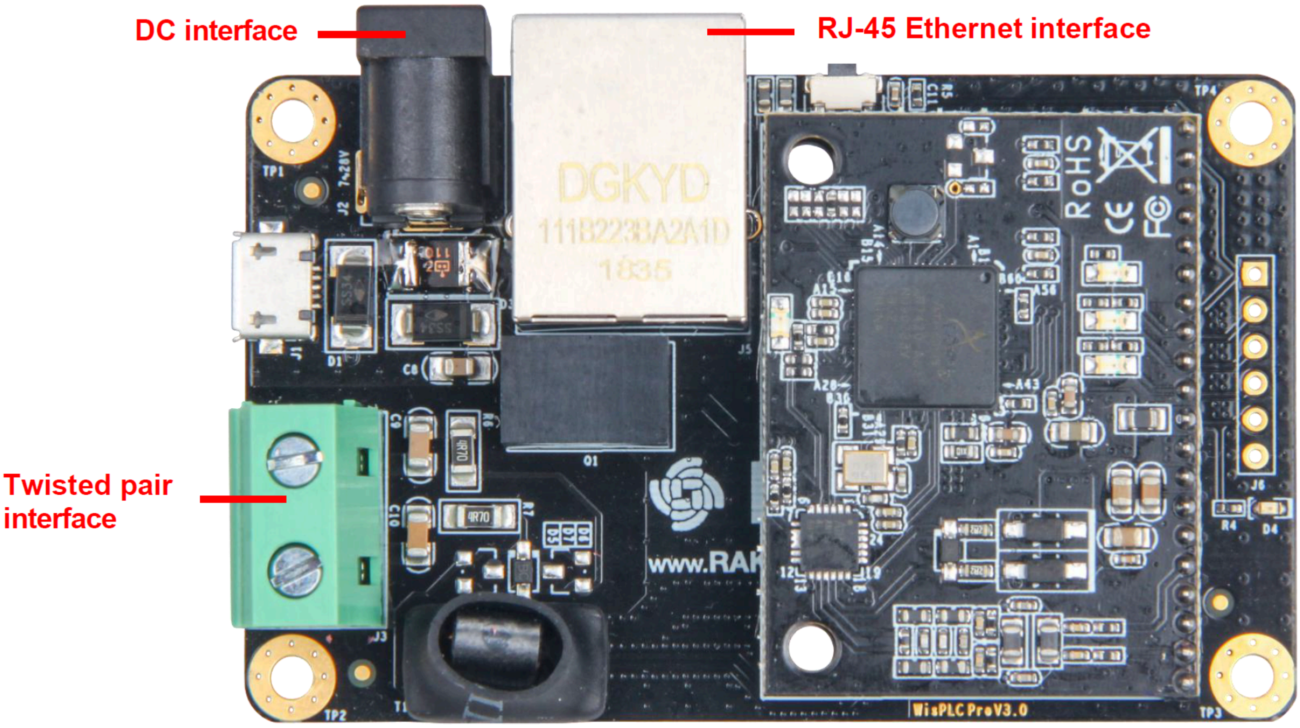

Figure 2: Baseplate interfaces and top view

Pin Definition

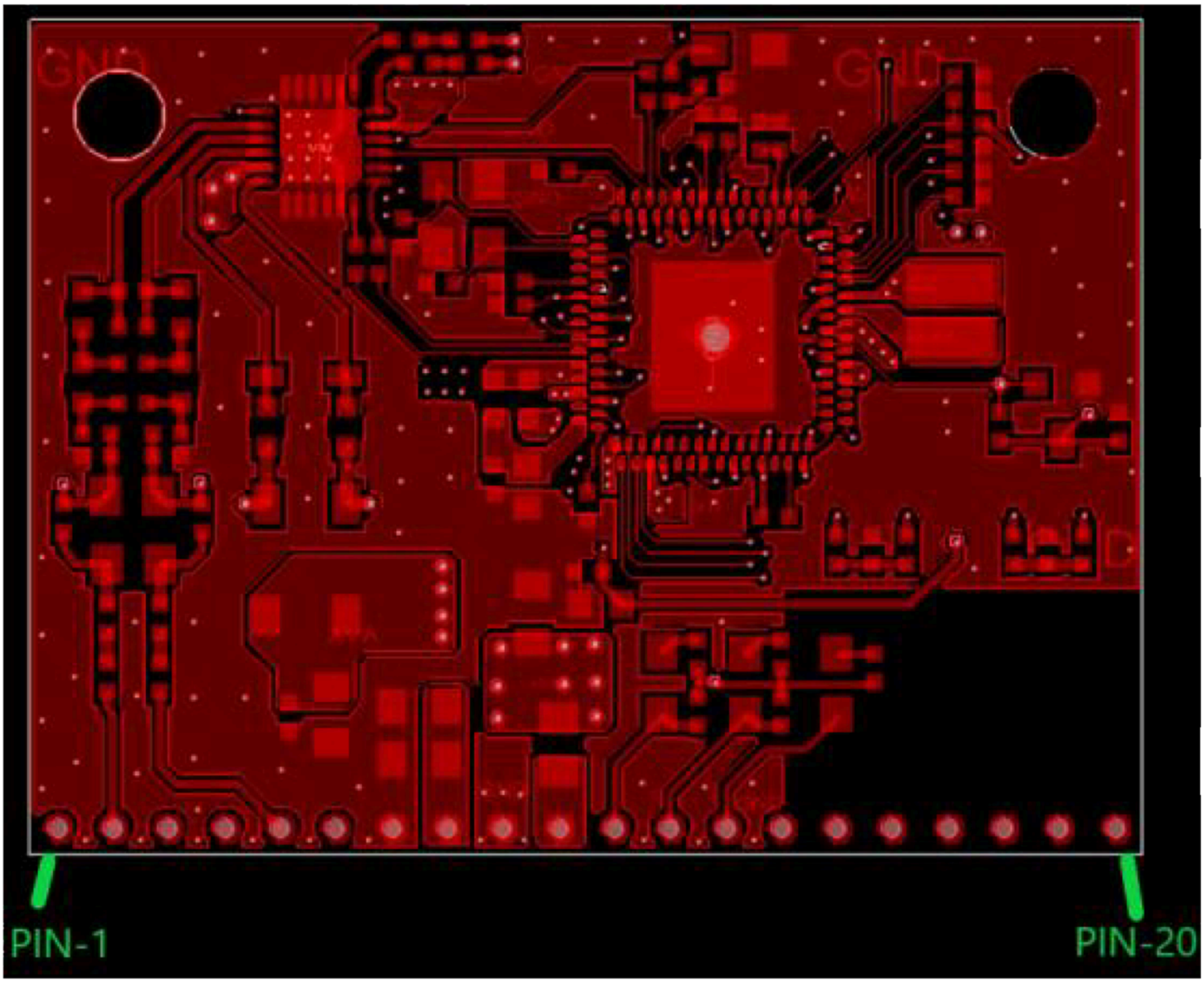

Figure 3: Pin definition

Pin No.	Definition	Description
1	NC1	NC
2	TR1-	Coupling signal sending, negative
3	NC2	NC
4	NC3	NC
5	TR1+	Coupling signal receiving, positive

Pin No.	Definition	Description
6	NC4	NC
7	GND	Ground
8	IN 12 V	12 V input operating voltage
9	GND	Ground
10	IN 3.3 V	3.3 V input operating voltage
11	PWR	Power indicator
12	ETH	Net-port ACT light, flashing when there is data flow
13	PLC	Signal transmission indicator light
14	RST	Reset
15	ETX+	Network sending signal, positive
16	ETX-	Network sending signal, negative
17	ERX+	Network receiving signal, positive
18	ERX-	Network receiving signal, negative
19	NC5	NC
20	NC6	NC

Electrical Characteristics

Parameter	Min	Typical	Max	Unit
12 V operating voltage	11.15	12	12.85	V
3.3 V operating voltage	3.15	3.3	3.45	V
Operating current	125	130	145	mA

Mechanical Characteristics

Schematic Diagram

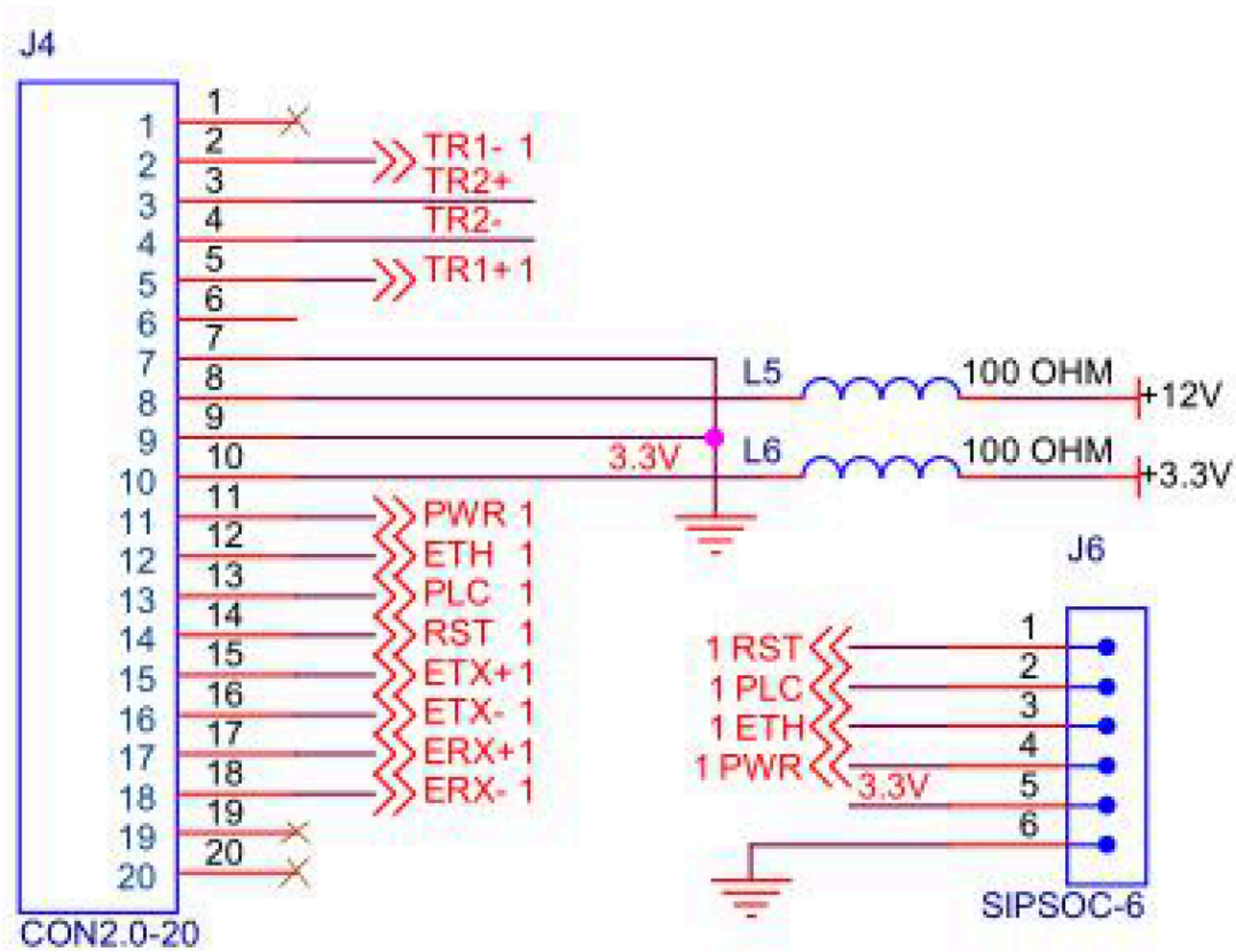

Figure 4: Connector pin map

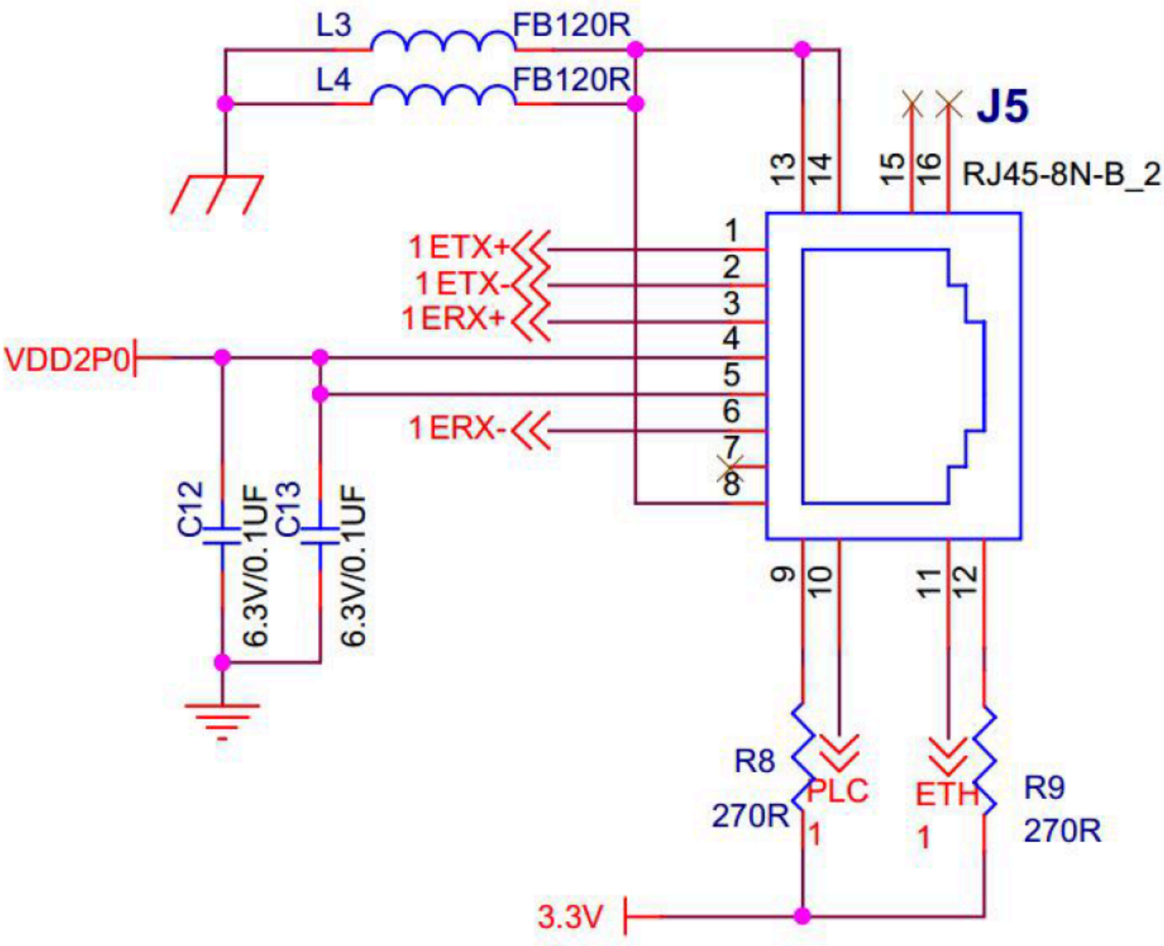

Figure 5: Ethernet port diagram

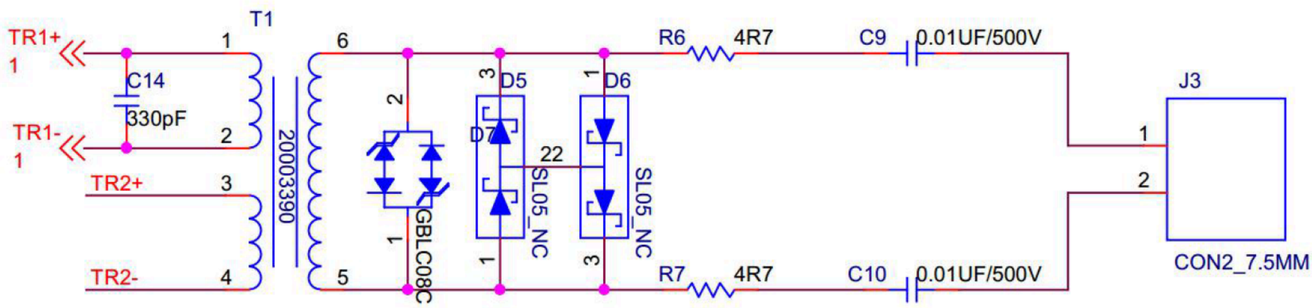

Figure 6: Signal receiving diagram

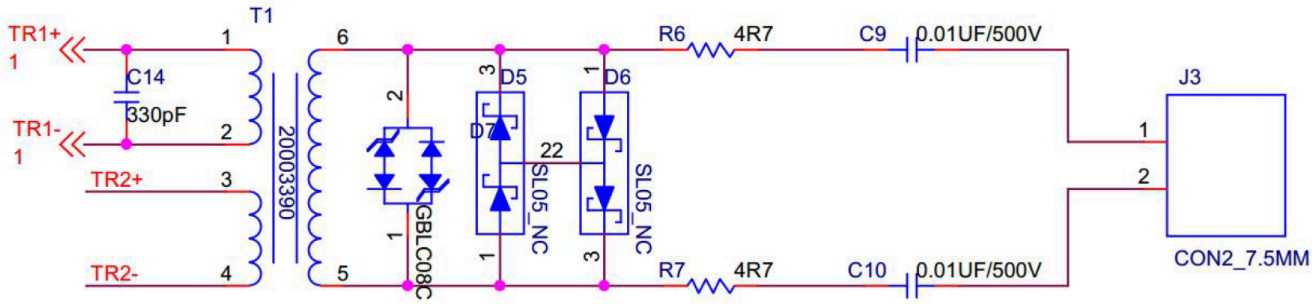

Figure 7: Signal indicators, factory parameter restoring circuit, and all kinds of filter circuits

Installation Guide

Assembly

The LX200V30 module is connected to the EVB via the pin connector. Simply slide the pins of the module in the connector of the board.

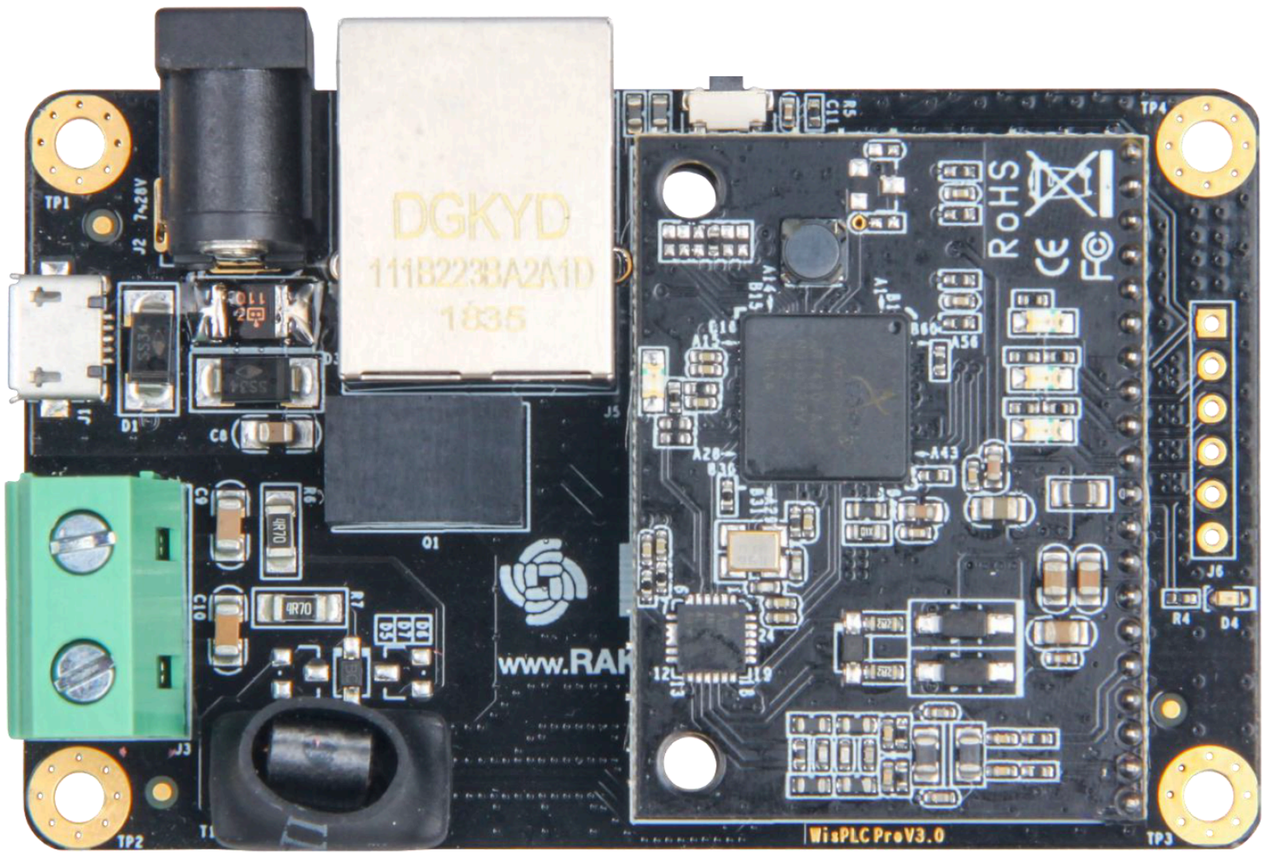

Figure 8: Assemble the module

NOTE

1. The signals RX+ and RX-, TX+ and TX-, PERX_P and PERX_N, PETX_P and PETX_N all use differential signal lines, and the cabling should follow the rules for differential signal.
2. The width of RX+ and RX-, TX+ and TX- should be no less than 20 mm. The length of the lines must not be more than 15 cm.